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Effects of module stiffness and initial compression on lithium-ion cell aging

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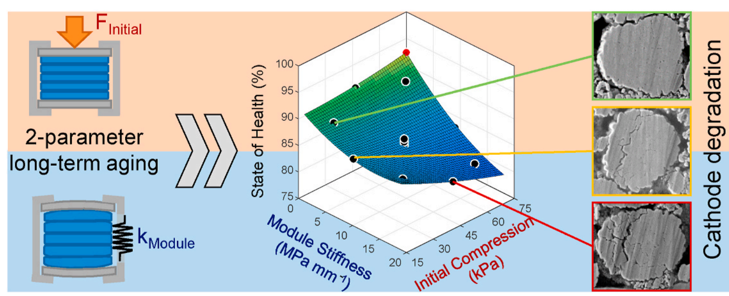
HIGHLIGHTS

- Cell swelling and aging under module stiffnesses and initial compressions
- Significant effects of both factors on cell aging and pressure evolution
- Pressure correlates with capacity fade due to loss of cathode active material
- Active material loss confirmed by half-cell measurements and SEM cross-sections
- Fatigue-based aging model of cathode particle cracking

GRAPHICAL ABSTRACT



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ABSTRACT

The effects of automotive-related lithium-ion module design, *i.e.* module stiffness and initial compression during module assembly on cell aging, swelling and pressure evolution are still largely unknown. This paper presents the results of a long-term aging study of 12 large-format automotive graphite/NMC 622 pouch cells, cycled for different module stiffnesses and initial compressions using design of experiments. Statistical analysis of mechanical and aging data revealed significant nonlinear (interaction) effects of both factors on pressure evolution, capacity loss and increase in internal resistance of the cells. Pressure dependent cell aging is observed over 1000 cycles, which was related to loss of active material at the cathode from differential voltage analysis. Post-mortem analysis confirmed a cathode active material loss via half- and full-cell measurements of harvested electrodes. Cross-section SEM micrographs revealed increasing NMC-particle cracking with higher pressure. Based on this, a fatigue-based aging model was developed to describe the capacity loss due to pressure dependent particle cracking. The presented approach enables both improved modeling of pressure dependent aging and lifetime optimized module design

1. Introduction

Aging of lithium-ion batteries is becoming increasingly important, as batteries need to endure several thousand charging and discharging cycles and many years of storage for their use in plug-in hybrid and electric vehicles. Consequently, it is imperative to understand cell aging and the underlying mechanisms to improve the lifetime of a battery. However, the interactions between automotive module design and cell aging are still largely unknown. Especially the mechanical bracing of cells in modules affects the pressure evolution and consequently both lifetime and safety, as modules have to withstand these dynamic loads without mechanical failure.

Lithium-ion cells undergo significant volume changes due to

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Nomenclature

a, b, c	Fitting parameters of the aging model
b_0	Constant term
b_1	Coefficient of the linear effect of module stiffness
b_2	Coefficient of the linear effect of initial compression
b_3	Coefficient of the interaction effect
b_4	Coefficient of the quadratic effect of module stiffness
b_5	Coefficient of the quadratic effect of initial compression
D	Accumulated damage
IC	Linear effect of initial compression
IC^2	Quadratic effect of initial compression
k	Number of independent stress events
m	Slope of the Wöhler curve
MS	Linear effect of module stiffness
MS^2	Quadratic effect of module stiffness
$N_{\max}$	Number of cycles to failure
n_i	Number of load cycles
SOH	State of health
y_i	Mean response
ε_i	Error term
σ_{ampl}	Stress amplitude
σ_{max}	Maximum stress
σ_{min}	Minimum stress
σ_{yield}	Yield strength

lithiation induced phase transitions of the active materials. Reversible volume changes of up to 10% have been reported for graphite [1,2] and 1%–5% for layered oxide cathodes such as $Li(Ni_xMn_yCo_z)O_2$ (NMC), depending on the Ni content [3,4]. Additionally, multiple aging mechanisms cause irreversible cell swelling. The formation and growth of the Solid-Electrolyte-Interphase (SEI) lead to an increase in particle size and expansion of the porous network of the anode [5–8], accompanied by a loss of active lithium and capacity. Lithium plating increases the cell thickness [9–11] and preferably occurs during high current charging, at high state of charge (SoC) and at low temperatures. Gas evolution can cause significant cell swelling, especially for prismatic hardcase cells [12,13]. In pouch-bag type cells, gas usually gets expelled from the electrode stack into the gas pockets when a sufficient external pressure is applied [14].

In automotive modules, multiple lithium-ion pouch cells are stacked and constrained by a rigid module housing. When the cells expand during charging, the pressure in the module rises [7,15–18], as the module frame is under tension and the cells are increasingly compressed. Irreversible cell swelling leads to increasing pressure in the module and onto the cell stack. Pressure has been shown to affect cell performance, as the ion transport is hindered under compression leading to an increase in ionic resistance [18–21]. Inhomogeneous pressure can lead to local lithium plating due to separator pore closure and heterogeneous current distribution [20–22]. However, applying a low pressure improves the electrical contact of particles [23], prevents delamination of electrode layers [15] and expels gas from the electrode stack [14,24].

Module design, *i.e.* the module stiffness and the initial compression during the module assembly process directly affects the resulting pressure [25,26] and therefore cell performance and aging. Increasing the initial compression leads to accelerated capacity fade of wound graphite/$LiCoO_2$ (LCO) cells [15], stacked graphite/NMC cells [27] and Si–C/NMC 811 cells [19]. This was mainly attributed to lithium plating and inhomogeneous lithium distribution [15,28] due to separator pore closure [15,19,28] and local deactivation of active material [19]. Sutter et al. [29] determined a low pressure of 31 kPa as optimal initial compression for enhanced capacity retention of Si-Alloy/NMC 622 cells. However, the results of Holland [30] show no effect of initial compression on aging of stacked graphite/NMC cells up to a pressure of 0.8 MPa. Barai et al. [23] reported almost identical aging for two different initial compressions, but also readjusted the pressure every 150 cycles. So far, the effects of module stiffness have been mostly neglected in literature. Wünsch et al. [31] showed a slightly lower capacity loss of a graphite/NMC 622 cell cycled with buffer elements compared to a rigid braced cell. Li et al. [13] investigated the influence of the gap between electrodes and the prismatic cell housing on cell aging and swelling force both on cell and module level. A smaller gap lead to a significantly higher swelling force and capacity loss. However, when comparing the module and cell data, it becomes obvious that it is imperative to replicate the exact mechanical module conditions on cell level to ensure comparability of the results.

To the authors knowledge no experimental study has so far studied the influence of initial compression and module stiffness on cell aging, swelling and pressure evolution. In order to evaluate effects from both factors and possible nonlinear interaction effects, an aging study is conducted, employing design of experiments. Furthermore, the pressure dependent aging mechanisms are verified by post-mortem analysis in order to develop a semi-empirical aging model.

2. Experimental

2.1. Design of experiments

The aim of this aging study is to examine the influence of module stiffness and initial compression on pressure evolution, irreversible cell swelling and aging of cells using design of experiments. In order to evaluate and optimize possible nonlinear effects of both factors and interaction effects, a central composite design (CCD) was chosen [32]. This experimental design is shown in Fig. 1a, where each point represents an individual experiment, respectively four replications at the central point, with the corresponding factor level combination.

All factor level combinations are listed in Table 1 with the respective cell number. The values for initial compression were derived from literature [26,28–30,33] and are in the range of typical values given by cell manufacturers to ensure optimal lifetime. Initial compression was carried out at 30% SoC, as modules are usually braced with this SoC due to transport safety regulations. The module stiffnesses were derived from literature [31,34,35] and state-of-the-art automotive battery modules.

2.2. Investigated cells

In this study, 12 long-format pouch cells with a graphite anode and a NMC 622 cathode were investigated. The 60 Ah high-energy cells consist of 34 stacked anode sheets and 33 cathode sheets with a stack length and width of 322 and 97 mm, respectively. Two gas pockets are located at the opposing tabs. According to the cell manufacturer, the cells can be operated between 2.8 V and 4.21 V. All cells were delivered and braced with a SoC of 30%.

2.3. Test setup and mechanical parameters

Cell bracing and cycling was conducted in our recently developed cell press (*cf.* Fig. 1b), achieving well controllable mechanical conditions. Due to the active regulation, it is possible to simulate any desired module stiffness and initial bracing process on cell level. The force acting on the cell is regulated based on its expansion and corresponds to the reaction force in a real module. Force and cell thickness are measured with high precision and logged time-synchronously with the

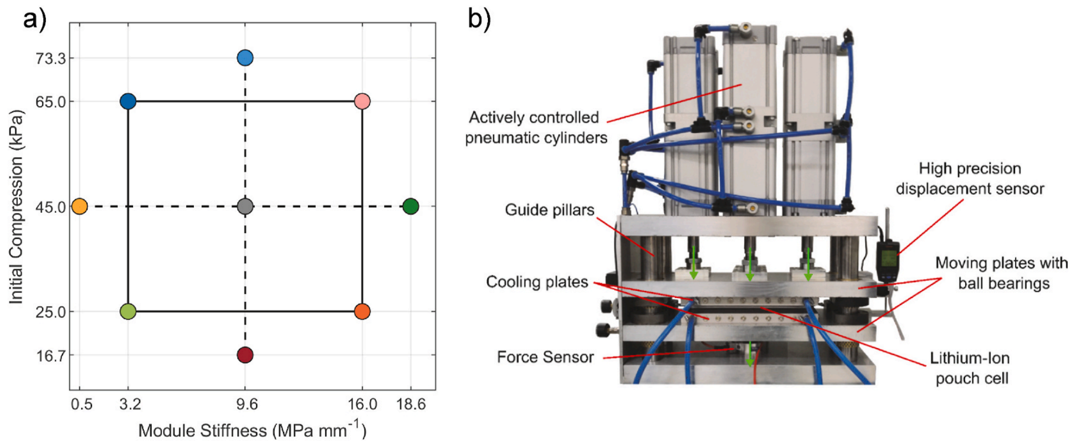

Fig. 1. a) Central composite experimental design for two factors: module stiffness and initial compression on five levels. b) Actively controlled pneumatic test setup for mechanical representative cell aging with high precision force and displacement measurement.

Table 1
List of all experiments, cell numbers and factor level combinations.

Cell	Module Stiffness (MPa mm^{-1})	Initial Compression (kPa)
1	3.2	25
2	16.0	25
3	3.2	65
4	16.0	65
5	0.55	45
6	18.6	45
7	9.6	16.7
8	9.6	73.3
9–12	9.6	45

electrical cycling data [25].

The cells were placed centrally in the cell press in order to achieve a homogenous pressure on the electrode stack without applying any pressure on the gas pockets. The cell press was closed slowly until a rise in force (>100 N) was detected and the force was then regulated to 100 N (3 kPa) for 5 min. Afterwards, the force was ramped up to the defined initial compression within 2 min. The pressure was held constant for 30 min before measuring the initial cell thickness and starting the module stiffness regulation. A subsequent pause of 1 h allowed for a setting of the force. The module stiffness regulation remained active for the complete cyclic aging experiment, regulating the force based on the reversible and irreversible cell swelling.

2.4. Cyclic aging experiments

The cells were charged and discharged for 1000 full cycles using a step-charging protocol provided by the manufacturer to prevent lithium plating (*cf.* Fig. S1). The voltage-based charging protocol consists of multiple constant current (CC) and constant voltage (CV) phases up to 4.21 V ($I_{cutoff} = C/20$). The cells were discharged with 1C CC to 2.8 V. Subsequent to charging and discharging a rest period of 15 min was assured.

At beginning of life (BoL) and after every 100 cycles, a check-up was carried out to determine capacity loss and internal resistance of the cells. During check-up, the cells were charged with the step-charging protocol and discharged with 1C CC-CV to 3.3 V ($I_{cutoff} = C/20$) three times. After a full charge and an extended pause of 1 h, the cell was discharged with C/10 to allow for a differential voltage (DV) analysis. A discharge pulse (100 A, 30 s) was carried out at 100% SoC to measure the internal resistance. Due to a malfunction of the battery cycling system, the second check-up for cells 2 and 7 was performed after 42 and 120 cycles, respectively. The following check-up was conducted after 200 cycles for both cells and every 100 cycles afterwards.

All cells were tempered to 25 °C with cooling plates connected to lab-cooling devices. Both cooling capacity and homogeneity were successfully validated in our previous work [25].

2.5. Cell volume measurement

In order to measure the volume change due to gas formation, a cell volume measurement was performed at the end of the test according to the Archimedes principle, where the cells buoyant force is derived from the difference of weight force and measured force when submerged completely in deionized water [36,37]. The force measurement was conducted with a high precision force sensor (FH10, Sauter, Germany). Comparing the measured volume to the volume of a fresh cell, discharged to the same cell voltage, enables an evaluation of cell volume change, as the cell volume differences are negligible at BoL.

2.6. Post-mortem analysis

A post-mortem analysis was performed on four aged cells and a fresh cell. Three cells (Cells 1, 2 and 6) and the fresh cell were discharged to 2.8 V ($I < C/100$) prior to the opening in an argon-filled glove box (MB200MOD, MBraun Inert-Gas Systeme, Germany) to minimize safety risks and ensure the same SoC for all cells. The pouch foil and both electrode tabs were cut with a ceramic knife and the electrode and separator sheets were removed one by one. The sheets were subsequently soaked in dimethyl carbonate for 4 min to remove residues of the electrolyte's conducting salt. After drying, the sheets were prepared for their use in three-electrode cells and further analyses [36]. Cell 4 was opened at 100% SoC at atmosphere in order to evaluate the homogeneity of lithium distribution in the anode. To restore the diffusion state weeks after the end of the aging experiment, the cell was cycled again for 29 cycles with the same cycling protocol in a cell press under constant pressure at 25 °C. Within 5 min after being fully charged, the cell was placed in a container filled with dry ice to slow down the diffusion within the electrodes. This has been shown to cause no additional damage to the cell [38].

Three-electrode full- and half-cells (ECC-PAT-Core, EL-Cell, Germany) were built from the extracted electrode layers. One side of the anode and cathode coating was carefully removed with 2-propanol and n-methyl-pyrrolidon, respectively. 18 mm diameter electrodes were extracted using a high precision cutting tool (EL-Cut, EL-Cell, Germany). A lithium foil with a diameter of 18 mm was used as counter-electrode for all half-cells. The separator was soaked with 110 µl and 95 µl of electrolyte for the half- and full-cells, respectively. The electrolyte was provided by the cell manufacturer and consists of EC/EMC (3:7) and $LiPF_6$ as conducting salt. Half- and full-cells underwent formation with multiple charging and discharging steps. Subsequently, full and half-cell capacities were obtained from a C/20 ($C_{full\text{-}cell} = 7.5$ mA, $C_{cathode} = 7.5$ mA, $C_{anode} = 8.1$ mA) discharge. The full-cells were cycled within the voltage limits provided by the manufacturer (2.8 V - 4.21 V). Anode half-cells were cycled between 0.03 V and 1 V and cathode half-cells between 3.3 V and 4.3 V.

Samples from one anode and cathode layer of each cell were extracted and analyzed by top view scanning electron microscopy (SEM). One sample was cut with an argon ion beam in order to analyze the microstructure of the electrode in cross-section SEM. The cathode side view was processed with ImageJ [39] to count and measure particles using a watershed segmentation algorithm and careful segmentation by hand. The anode cross section micrographs were segmented using a Weka Segmentation Algorithm [40], that was trained to identify macroscopic and microscopic porosity of the electrodes.

Inductively coupled plasma optical emission spectrometry (ICP-OES) was used to investigate the composition of samples from cathodes of a fresh cell and aged cell 6. Five samples of the unilaterally decoated electrode with 18 mm diameter were extracted from each cathode and charged with multiple CC-CV steps to 4.2 V vs. Li/Li^+ in half-cells using lithium as counter-electrode. After washing and drying, the electrodes were dissolved in aqua regia and subsequently analyzed with an Agilent 5100 ICP-OES (Agilent Technologies, U.S.A.). Additionally, eight samples with 18 mm diameter were extracted from the anodes of a fresh cell, cells 1, 2 and 6. After dissolving in aqua regia, the samples were analyzed with a Vista Pro (Varian, U.S.A.) ICP-OES system.

3. Results and discussion

3.1. Aging, pressure evolution and cell swelling

Aging of lithium-ion cells is characterized by the fade of capacity and power measured in the check-up tests. The evolution of the state of health (SoH), *i.e.* the C/10 discharge capacity relative to the initial capacity, is depicted for all cells in Fig. 2a. The initial capacity of 59.0 ± 0.2 Ah at the first check-up indicates only small variance between the cells. Afterwards, all cells except for cell 4 (16 MPa mm^{-1}, 65 kPa) show a mostly linear capacity loss. In the first 500 cycles, cell 4 exhibits a fast linear capacity loss that flattens subsequently. This might be linked to the pressure limitation of the cell press. Nevertheless, cell 4 shows the highest capacity loss with a SoH of 81.2% at the end of test (EoT). In contrast, cell 5 (0.55 MPa mm^{-1}, 45 kPa) has a remaining capacity of 91.1% after 1000 cycles, which marks the lowest capacity loss. Cells 9–12 (3.2 MPa mm^{-1}, 45 kPa), the four replications from the center

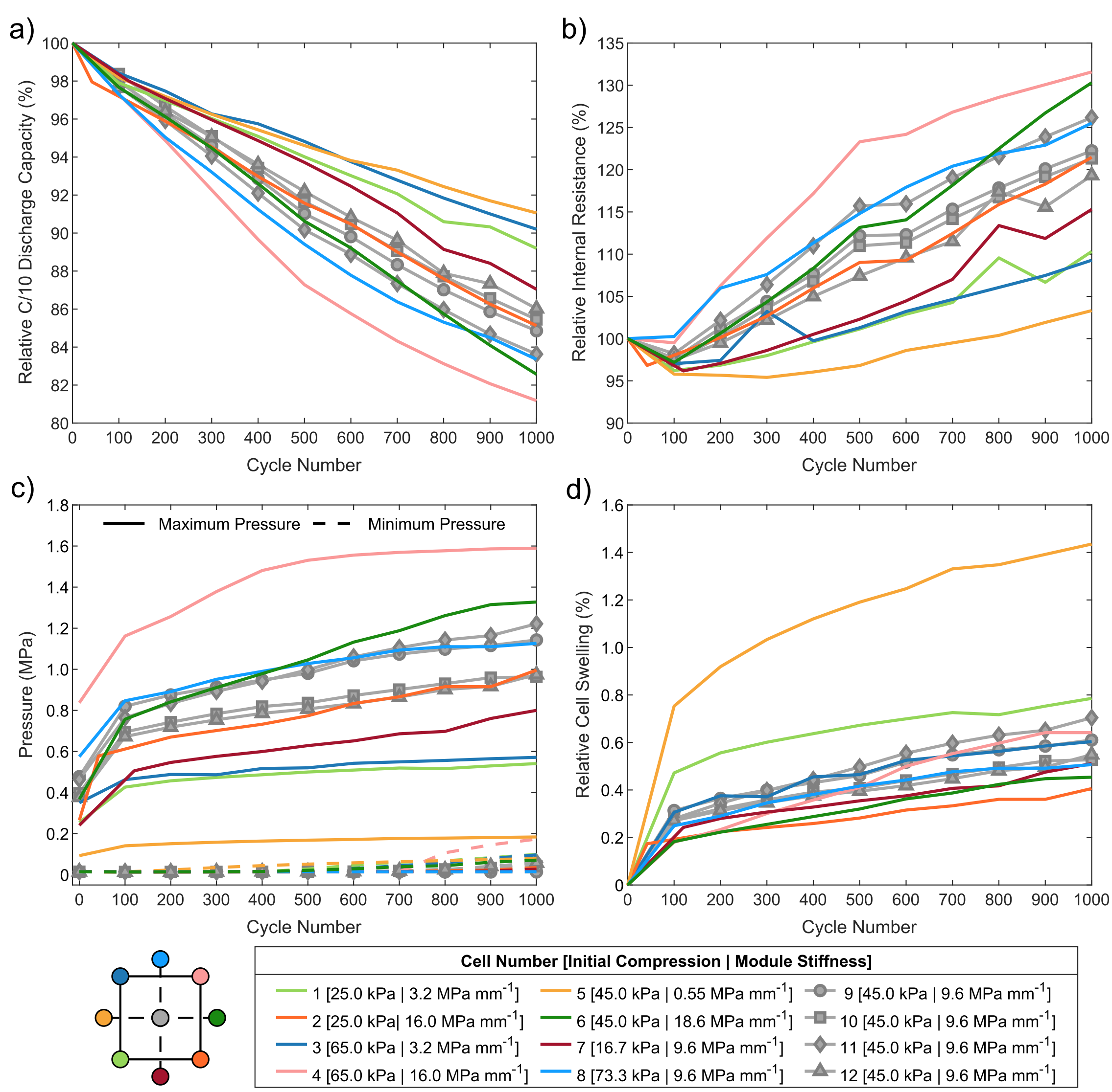

Fig. 2. Aging results of all experiments over 1000 cycles: a) Relative capacity from C/10 discharge (C = 60 A), b) relative internal resistance from 100 A discharge pulse at 100% SoC, c) maximum and minimum pressure and d) relative cell swelling at 100% SoC.

point, show a spread of 83.6%–86.0% in their SoH, which could have been partly caused by a variance of process parameters during cell production [41]. A possible correlation to differences in pressure evolution will be discussed below.

An overview of the evolution of internal resistance is pictured in Fig. 2b. Initially, the internal resistance equals 1.59 ± 0.02 mOhm independent of the bracing conditions. The internal resistance of most cells stays constant or even decreases for at least 100 cycles. After cycling, the internal resistance of cell 4, which exhibited the highest capacity fade, increases by 31.5%. At the same time, cell 5 displays only a minor increase of 3.3%, correlating with the lowest capacity fade. Again, the variance of internal resistance increase can be estimated by the four replicated measurements (cells 9–12), which lie in a range from 19.3% to 26.2% at EoT. Within this range, higher increases in internal resistance correlate with higher capacity losses.

The evolution of both maximum and minimum pressure during check-ups is shown in Fig. 2c, where again, cells 4 and 5 mark the extreme points. The combination of a high module stiffness and high initial compression of cell 4 leads initially to a maximum pressure of 0.84 MPa when fully charged. After a strong initial pressure growth in the first 100 cycles, the growth rate starts to decrease resulting in a maximum pressure of 1.59 MPa at EoT, which is limited by the cell press. The coincidentally decreasing capacity loss rate might indicate a pressure dependent aging mechanism that slows down rather due to the limitation of the cell press than due to a passivating effect. The minimum pressure when fully discharged reaches the lowest possible pressure of 0.01 MPa for 700 cycles. In a real module, the pressure would decrease to ambient pressure, as the cells would shrink out of the module. In order to measure the cell thickness below this point, a small constant pressure is applied [25]. Afterwards, the cells are always constrained by the module as the minimum pressure rises to 0.17 MPa. The lowest module stiffness and a medium initial compression lead to a maximum pressure of 0.09 MPa for cell 5 at BoL. The maximum pressure grows to 0.18 MPa during aging, while the minimum pressure starts to rise early after 100 cycles before reaching 0.08 MPa. The initial maximum pressure of cells 9–12 lies in the range of 0.37 MPa and 0.48 MPa. Assuming an identical volume expansion at BoL, this is most likely due to differences in the mechanical cell stiffness. During aging, this spread widens, resulting in maximum pressures in the range of 0.96 MPa–1.22 MPa. The reason for this might be a higher divergence of the non-linear cell-stiffnesses at higher pressures. The pressure evolution of these cells show the same trend as for the capacity loss and the internal resistance, which suggests a coupled aging mechanism.

The relative cell swelling in Fig. 2d denotes the irreversible cell swelling at 100% SoC compared to the initial cell thickness in the first cycle. All cells experience a strong initial cell swelling in the first 100 cycles that accounts for a great portion of the overall cell swelling (*cf.* Fig. 2d). As expected, cell 5 is able to swell nearly unconstrained up to 1.4% after 1000 cycles due to the low module stiffness. Cell 4 displays a relatively high cell swelling of 0.6% despite the high pressure acting on it. A range of 0.5%–0.7% is observed for the four replications, which directly corresponds to the differences in pressure evolution.

In order to gain a better understanding of the effects and interactions of both module stiffness and initial compression on the response variables, it is crucial to perform a statistical and subsequently a correlation analysis.

3.2. Statistical analysis

It is possible to create response surface models based on the CCD that estimates *i* mean responses (y_i) of linear and quadratic effects of module stiffness (MS) and initial compression (IC) using a quadratic function (eq. (1)) [32].

$$y_i = b_0 + b_1\,MS + b_2\,IC + b_3\,MS\,IC + b_4\,MS^2 + b_5\,IC^2 + \varepsilon_i \qquad \text{(eq. 1)}$$

In eq. (1) b_0 denotes a constant term, b_1-b_5 are the estimators for the coefficients corresponding to the factors and ε_i stands for the error term. The model is derived by fitting the coefficients to the dataset using linear regression and least squares method. In order to evaluate the statistical significance of the model terms, an analysis of variance (ANOVA) is performed for each of the response variables maximum pressure, SoH and internal resistance at every check-up. Furthermore, the optimal factor combination is determined by minimizing or maximizing the quadratic function depending on the response variable.

The model terms and the results of the ANOVA, the p-value (derived from the standardized factor levels) and the evaluation of significance, for maximum pressure at BoL and at EoT are summarized in Table 2. The statistical significance of each coefficient in the model is evaluated on three levels marked with asterisks. A p-value $\leq .05$ is rated statistically significant (*), ≤ 0.01 very significant and ≤ 0.001 extremely significant. At BoL all coefficients except for the quadratic terms were rated at least statistically significant. The complete model describes the maximum pressure with a low root–mean-square-error (RMSE) of 0.08 MPa and a R^2 of 0.91, indicating a high goodness of fit. The significantly lower value of R^2_{adj} that accounts for the number of fitting parameters, indicates slight overfitting due to the inclusion of the quadratic terms. In the last check-up, all previous model terms and the quadratic effect of module stiffness are at least statistically significant. The corresponding R^2 and R^2_{adj} reveal a good model fit without any sign of overfitting and a low RMSE of 0.10 MPa. The 3D surface and 2D contour plot of the response surface model at EoT are shown in Fig. 3a with the corresponding experimental results and the optimum marked in black and red, respectively. The main and interaction effects of module stiffness and initial compression are visible from both diagrams. An increase in module stiffness leads to a higher pressure. The effects of initial compression however, are mostly visible when applying high module stiffnesses. Here, the maximum pressure rises with increasing initial compression. The highest pressure of over 1.6 MPa can be reached by maximizing both initial compression and module stiffness. Conversely, the lowest pressure occurs at a low module stiffness and according to the model, at a high initial compression.

The effects of module stiffness and initial compression on maximum pressure can be explained from a mechanical viewpoint. Assuming no differences in cell stiffness, breathing and swelling, a higher module stiffness invokes a higher reaction force onto the cell. The initial compression defines the point of zero strain for the module bracing. With increasing initial compression, a higher percentage of cell breathing and cell swelling is absorbed from the module bracing, resulting in a higher overall pressure. Additionally, a higher equilibrium force is reached due to the nonlinearity of the cell stiffness that increases with rising pressure. Both trends agree with the model except for the optimum of pressure, which is predicted at a high initial compression (*cf.* Fig. 3a). One would expect the pressure to decrease by minimizing both module stiffness and initial compression. As the effects of the initial compression are smaller for lower module stiffnesses, this might be a model flaw. The reason for the spread at the central point is a variance of cell stiffness possibly caused by tolerance deviations during cell production. The results for the initial compression at high module stiffnesses are in agreement with the investigations of Cannarella et al. [15] and Holland [30], who reported a higher maximum pressure during cycling with increasing initial compression. The effect of module stiffness on pressure evolution until end of life (EoL) matches the prediction of Choi et al. [26], where cell aging in a high stiffness structure lead to significantly higher pressures compared to a low stiffness structure.

The results for the C/10 capacity at BoL and the SoH at EoT are summarized in Table 3. At BoL only the first coefficient in the model, the intercept, is statistically significant. Consequently, the model is not able to describe any differences in the measurements resulting in low values for R^2 and R^2_{adj}. However, the small RMSE of 0.20 Ah suggests a randomly distributed capacity, regardless of the module bracing. At the

Table 2
Results of the linear regression modeling of maximum pressure and the corresponding ANOVA at BoL and after 1000 cycles.

Coefficient	Model Term	Maximum pressure (BoL)			Maximum pressure (after 1000 cycles)		
		Estimate	p-Value	Significance	Estimate	p-Value	Significance
b_0	Intercept	2.63e-01	<.001	***	1.41e-01	<.001	***
b_1	MS	9.48e-03	.007	**	7.83e-02	<.001	***
b_2	IC	−5.44e-03	.002	**	5.20e-03	.010	**
b_3	MS*IC	9.30e-04	.023	*	1.10e-03	.034	*
b_4	MS^2	−1.77e-03	.057		−3.51e-03	.012	*
b_5	IC^2	4.05e-05	.619		−9.96e-05	.363	
R^2		0.91			0.96		
R^2_{adj}		0.83			0.93		
RMSE		0.08 MPa			0.10 MPa		

last check-up, every effect except the quadratic effect from the initial compression (IC^2) is at least statistically significant. Again, the goodness of fit is evaluated for the complete model with high values for R^2 and R^2_{adj}. The model is well suited to describe the differences in SoH with a small RMSE of 0.85%.

The resulting model for the SoH at EoT is depicted in Fig. 3b both as a response surface and contour plot. The lowest SoH (<80%) and therefore the highest capacity loss is reached for a high module stiffness and a high initial compression, correlating with the highest pressure (*cf.* Fig. 3a). Optimal aging with a SoH of more than 92% is predicted for minimizing module stiffness and maximizing initial compression, correlating with the lowest pressure evolution.

These drastic differences in aging can only be explained by the effects of the module bracing, *i.e.* by the pressure acting on the cells, as all cells were cycled under the same thermal and electrical load. In other words, optimizing the module design can lead to a significant improvement of lifetime, up to a factor of two in capacity fade. However, only a post-mortem analysis can provide insights into the exact mechanisms of capacity loss. The results for the initial compression at high module stiffnesses align with the results of Cannarella et al. [15], where an increase in initial compression lead to a higher capacity loss of wound pouch-cells. Similar dependencies for stacked pouch-cells were reported by Ebert et al. [27] and Müller et al. [19]. However, Holland [30] could not verify any effects of initial compression for stacked pouch-cells in a long-term aging study. The effects of module stiffness have a significant effect on capacity loss but have been neglected in literature so far.

The results for internal resistance closely resemble these for SoH and are presented in Table 4. At BoL no factor is deemed statistically significant apart from the intercept. Logically, this results in low values for R^2 and R^2_{adj}. The internal resistance at BoL is randomly distributed and not dependent on the module bracing. Similar to the SoH, this changes during aging and the linear terms and the quadratic term for the module stiffness are statistically significant at EoT. Both R^2 and R^2_{adj} show a good fit and no sign of overfitting, even for the complete model. The relatively high RMSE of 2.30% might stem from large cell-to-cell variations, which are visible at the central point.

The 3D surface and 2D contour plot of the final response surface model are shown in Fig. 3c. The highest increase in internal resistance (>135%) as well as the highest capacity fade and pressure evolution is predicted for a high module stiffness and high initial compression. Again, a low module stiffness and a high initial compression represents the optimum. As the capacity was measured with a low discharge current, the influence of internal resistance is assumed negligible and both measurements can be analyzed separately. The higher internal resistance with increasing initial compression is in agreement with the results of Cannarella [28], who reported this effect to be reversible as a removal of the applied pressure lead to a significant reduction of cell impedance. As the rise in internal resistance could stem from different aging mechanisms, a thorough post-mortem analysis is needed to explain this behavior.

A correlation between pressure and SoH as well as internal resistance seems obvious from these results, but in order to validate this, a correlation analysis was performed by calculating the Pearson correlation coefficient (*cf.* Table 5). The Pearson correlation coefficient has a range of −1 to +1, where −1 and +1 denote a perfect negative and positive linear correlation, respectively. The coefficient for maximum pressure and SoH decreases during aging and approaches the ideal value of −1. The same negative linear correlation is obvious when plotting both the SoH and the respective maximum pressure and fitting a linear function to the data (*cf.* Fig. 4a). The linear slope changes with increasing number of cycles, as the cells exhibit faster capacity fade with increasing pressure. Accelerated aging under pressure is known from literature and mostly attributed to separator deformation [15,19], which can lead to lithium plating [20,21,28].

The clearly positive correlation of pressure and internal resistance and its evolution during aging is shown in Fig. 4b and summarized in Table 5. After merely 200 cycles, the Pearson correlation coefficient reaches a value of 0.91 and further increases to a very high value of 0.98. Similar to the SoH correlation, the slope of the linear function changes with increasing number of cycles (*cf.* Fig. 4b), due to the disproportionate growth of internal resistance at high pressure. This is in agreement with literature, as pressure leads to progressive pore closure and therefore to an increase in ionic resistance [18,24].

A correlation analysis of pressure and relative cell swelling was conducted in Fig. 4c and Table 5, revealing a partially negative linear correlation. In fact, instead of a linear fit, a two-term power series model was fitted to the data to show the general passivating behavior. With increasing pressure, cell swelling is commonly reduced as the cell is increasingly compressed. A decreasing cell swelling under pressure was observed by Müller et al. [19] and Choi et al. [26] for both reversible and irreversible cell swelling.

When designing a module, the volumetric energy density is of great importance and shows an interesting evolution during aging (see Fig. 4c, Table 5). At BoL, a positive linear correlation prevails. Consequently, a module that exerts a higher pressure at BoL offers a higher volumetric energy density at the same time. In order to optimize the energy density at BoL, one might try to design a stiff module and increase the initial compression. However, the correlation changes to a highly negative linear correlation during aging. Thus, the initially optimized module now exhibits the lowest energy density. The initial advantage of the stiff and highly compressed module is purely mechanical, as the capacity is not affected from the bracing at BoL. After 400 cycles, the higher capacity fade at the elevated pressure outweighs the initial advantages from the suppressed cell swelling.

Statistical and correlation analysis has further confirmed that cell aging strongly depends on the external pressure, which is in turn defined by both module stiffness and initial compression. However, the pressure-dependent aging mechanisms cannot be determined from these analyses. Consequently, a post-mortem analysis was performed to identify the dominant aging mechanisms and to explain the differences in aging for cells cycled for different pressures.

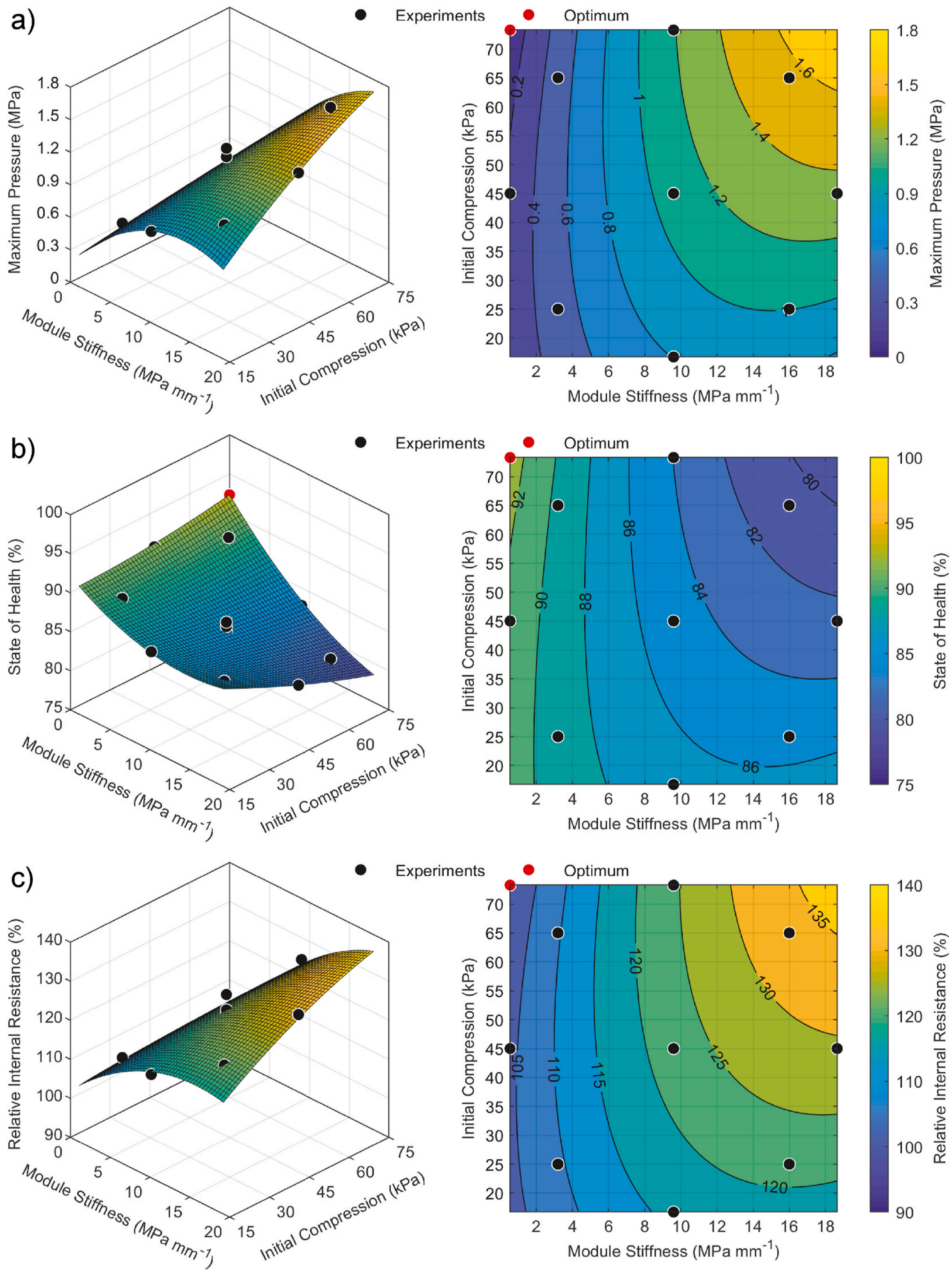

Fig. 3. Response surface and contour plots for a) maximum pressure, b) state of health and c) relative internal resistance after 1000 cycles plotted versus both factors module stiffness and initial compression. In addition to the experimental data (black), the optimum values are marked as red dots. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.3. Post-mortem analysis

The results of the post-mortem analysis and the most important cell parameters of the fresh cell (BoL), cell 1 (SoH 89.6%), cell 2 (SoH 85.5%), cell 4 (SoH 81.2%) and cell 6 (SoH 83.3%) are summarized in Table 6. Cell volume measurement revealed a negligible volume change (<10 ml), indicating that gas production had no influence on pressure evolution and aging mechanisms (*cf.* Table 6).

The electrode state diagrams derived from three-electrode full-cell measurements of the harvested electrodes are depicted in Fig. 6 on the left side. Anode and cathode potentials as well as the cell voltage over areal capacity during the C/20 discharge is displayed with a detailed view of the anode potential at the end of discharge. All three measurements of the fresh cell (*cf.* Fig. 5a) are highly reproducible with only slight deviations of the potential curves and capacity. It should be noted that the lower voltage limit is reached due to the increasing anode potential. Some trends are visible in the electrode state diagrams that intensify with increasing pressure and decreasing SoH (from top to bottom), when comparing cells 1, 2 and 6 (*cf.* Fig. 5c, e, g). The cells exhibit a worsening degree of inhomogeneity with increasing deviations

Table 3
Results of the linear regression modeling of capacity and SoH with the corresponding ANOVA at BoL and after 1000 cycles.

Coefficient	Model Term	C/10 Discharge Capacity (BoL)			State of health (after 1000 cycles)		
		Estimate	p-Value	Significance	Estimate	p-Value	Significance
b_0	Intercept	5.93e+01	<.001	***	9.12 E+01	<.001	***
b_1	MS	2.12e-03	.481		−5.32E-01	<.001	***
b_2	IC	−1.27e-02	.996		−5.06E-03	.014	*
b_3	MS*IC	5.54e-04	.502		−9.68E-03	.027	*
b_4	MS^2	−9.81e-04	.627		2.48E-02	.023	*
b_5	IC^2	8.28e-05	.687		5.22E-04	.556	
R^2		0.21			0.96		
R^2_{adj}		−0.44			0.92		
RMSE		0.20 Ah			0.85%		

Table 4
Results of the linear regression modeling of internal resistance with the corresponding ANOVA at BoL and after 1000 cycles.

Coefficient	Model Term	Absolute internal resistance (BoL)			Relative internal resistance (after 1000 cycles)		
		Estimate	p-Value	Significance	Estimate	p-Value	Significance
b_0	Intercept	1.66e+00	<.001	***	1.00e+02	<.001	***
b_1	MS	−4.30e-03	.613		1.75e+00	<.001	***
b_2	IC	−2.45e-03	.429		1.72e-01	.011	*
b_3	MS*IC	1.14e-04	.080		2.18e-02	.052	
b_4	MS^2	−2.25e-05	.873		−6.95e-02	.020	*
b_5	IC^2	1.74e-05	.253		−2.60e-03	.295	
R^2		0.55			0.96		
R^2_{adj}		0.17			0.93		
RMSE		0.01 mOhm			2.30%		

Table 5
Summary of Pearson correlation coefficients for maximum pressure and SoH, internal resistance, relative cell swelling as well as volumetric energy density at every second check-up.

	Maximum pressure					
	BoL	200 cycles	400 cycles	600 cycles	800 cycles	1000 cycles
State of health	-	−0.85	−0.93	−0.95	−0.97	−0.98
Internal resistance	-	0.91	0.97	0.97	0.98	0.98
Relative cell swelling	-	−0.75	−0.67	−0.79	−0.59	−0.64
Energy density	0.55	0.08	−0.70	−0.83	−0.91	−0.93

between the individual measurements with respect to capacity and half-cell potentials. With increasing pressure, the lower voltage limit is reached more frequently due to the sharp drop in cathode potential and a shift in electrode balancing between anode and cathode (*cf*. Fig. 5c, e and g). These two trends suggest a worsening cathode active material loss that is also becoming increasingly inhomogeneous.

The right side shows the constructed DV curves of the fresh cell and the evolution of DV curves during aging for the other cells corresponding to the electrode state diagrams on the left. The half- and full-cell DV curves of the fresh cell in Fig. 5b clearly show two peaks in both the anode potential and the full-cell curve, associated with the phase transitions of graphite during lithiation [42,43]. The NMC 622 has no distinct peaks but an increasing slope starting at ~40% SoC [44] with a further increase from 75% SoC on. The progressive cathode active material loss with increasing pressure changes the first increasing slope of the DV curves, associated with the NMC. The onset shifts to a lower SoC while the slope itself decreases, suggesting an inhomogeneous active material loss of the NMC. The second steep slope at higher SoC further increases with growing pressure during aging and the distance between the onset of the increasing slope and the end of discharge diminishes. These effects indicate cathode active material loss as the dominant aging effect. All cells show signs of heterogeneous lithium distribution in the DV curves as the graphite peaks flatten during aging (*cf*. Fig. 5d, f, h) [44,45].

All of these effects intensify with increasing pressure, suggesting inhomogeneous cathode degradation as the dominant aging mechanism for all cells. There is inevitably some lithium loss caused by SEI and Cathode-Electrolyte-Interphase (CEI) growth, but due to the flattening and later disappearing of the anode peaks, a quantification is not reasonable.

Areal capacities of the half-cells were obtained from harvested electrodes in order to further validate the active material loss (*c.f.* Fig. 6a and Table 6). While the anode capacity slightly increases by 1%–3%, the cathode capacity decreases by 5.9%–8.8%, indicating cathode active material loss. An increase in anode capacity has been observed in literature for state of the art cells [45]. A possible explanation for this could be a poor initial electrical connection of particles, that enhances during cycling due to structural changes of the anode [45] or the applied pressure. Any active material loss of the cathode diminishes the full-cell capacity due to its limiting electrode loading. Multiple aging mechanisms have been proposed for Ni-rich NMC cathodes in literature. The cathode active material loss could have been caused by a mechanical degradation due to the anisotropic volume expansion of NMC [46–48]. This can lead to intragranular micro-cracks within the particle [46,47] and subsequently to particle cracking and the pulverization of active material [46]. Additionally, electrolyte is able to penetrate the cracks and can be decomposed further leading to lithium loss [46–48]. Another explanation is chemical degradation, *i.e.* a reconstruction of the cathode surface or a phase transition to a NiO-like layer [49–52], which might be accelerated by a mechanical degradation. However, the observed cathode active material cannot explain the cell capacity loss solely. In order to further examine a possible lithium loss in the cathode, the lithium content of delithiated samples from a fresh cell and cell 6 was analyzed via ICP-OES. A significantly higher lithium content was found in the aged cathode (21.55 ± 0.13 g kg^{-1}) compared to the fresh cathode (20.18 ± 0.15 g kg^{-1}), indicating electronic isolation of lithiated particles that lead to a loss of cyclable lithium. As cell swelling usually

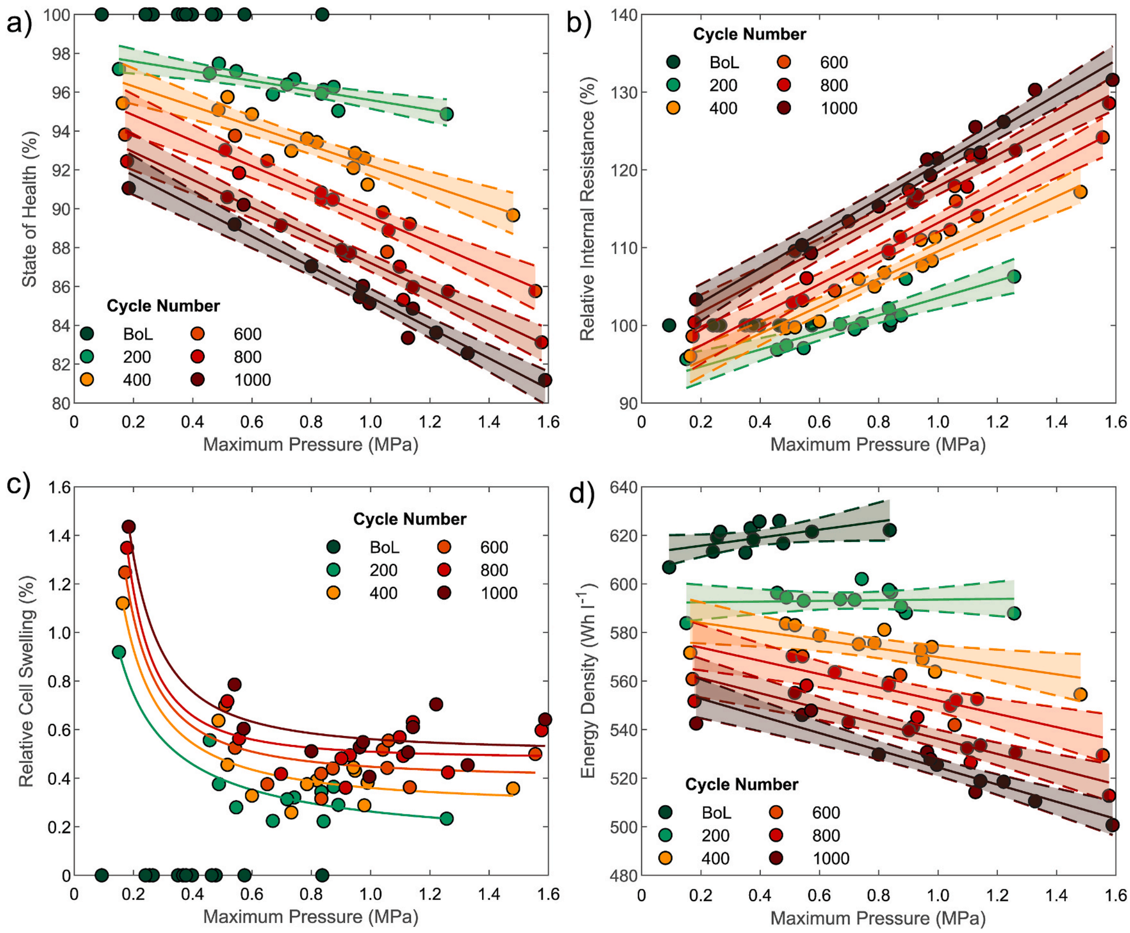

Fig. 4. Correlation of a) state of health, b) relative internal resistance, c) cell swelling and d) volumetric energy density with maximum pressure at BoL and every 200 cycles until the end of test. For a), b) and d) a linear fit and the corresponding confidence intervals are depicted. The relative cell swelling is fitted using a power law function and shown without the confidence interval for better readability.

Table 6
Summary of cell parameters and post-mortem analysis results: cell volume and thickness of the double-sided coated electrodes, half-cell capacities, anode porosity, cathode particle count and size.

Parameter	BoL	Cell 1	Cell 2	Cell 6
SoH (%)	100	89.6	85.5	83.3
Maximum pressure (MPa)	-	0.54	1.00	1.33
Cell volume change (ml)	-	9.2	4.1	2.6
Anode capacity (mAh cm^{-2})	3.479 ± 0.009	3.512 ± 0.018	3.584 ± 0.051	3.533 ± 0.016
Anode thickness (µm)	169.5 ± 1.0	174.7 ± 1.6	175.3 ± 0.9	179.4 ± 7.1
Anode porosity (%)	17.5	16.1	15.6	17.8
Anode Li mass fraction (g kg^{-1})	5.69 ± 0.10	9.41 ± 0.27	10.71 ± 0.20	10.99 ± 0.20
Cathode capacity (mAh cm^{-2})	3.272 ± 0.022	3.079 ± 0.082	3.057 ± 0.037	2.983 ± 0.0271
Cathode thickness (µm)	139.0 ± 1.4	140.5 ± 0.8	139.3 ± 0.8	136.8 ± 0.7
Uncropped cathode particles (–)	2919	3058	3052	3651
Median cathode particle size (µm^2)	0.300	0.292	0.292	0.244
Cathode Li mass fraction (g kg^{-1})	20.18 ± 0.15	-	-	21.55 ± 0.13

correlates with active lithium loss due to SEI growth [5–8,16] or lithium plating [9–11], a significant amount of lithium loss is expected from cell swelling. This is confirmed by the increasing lithium content in the anode (see Table 6) measured via ICP-OES. The higher lithium content in the anode of cell 2 compared to cell 1 might explain the observed differences in pouch-cell SoH (*cf.* Table 6). Here, pressure might have accelerated the cracking and growth of the SEI or promoted lithium plating.

In fact, the aged anodes show a 3.0%–5.9% increase in thickness compared to a fresh cell (*cf.* Fig. 6b). The increase is mostly uniform for cells 1 and 2, while cell 6 exhibits higher thickness variation (see Table 6). The observed anode growth is comparable to the range of 4.5%–14.3% reported in literature [44,53,54]. The higher anode thickness of cell 6 might be due to local lithium plating, as only two areas of the anode exhibit stronger growth than the other aged anodes.

The cathode thickness increases by 1.1% for cell 1, while cell 2 shows no increase at all. Even a slight decrease in cathode thickness was observed for cell 6. The evolution of cathode thickness during aging is still unclear from literature. Some researchers reported no irreversible thickness changes [54,55] for NMC/LMO blend cathodes. However, NMC 111 and NMC 622 cathodes have been shown to irreversibly grow 2.2%–9% during cycling [44,56]. In our investigation, the cathode of cell 1 cycled under the lowest pressure shows an increase in thickness while the thickness decreases for cell 6 under the highest pressure. This could indicate a plastic deformation of the cathode effectively suppressing cathode growth.

In order to validate morphological changes of the electrodes, the SEM cross-section micrographs (*cf.* Fig. 7, left and middle column) were

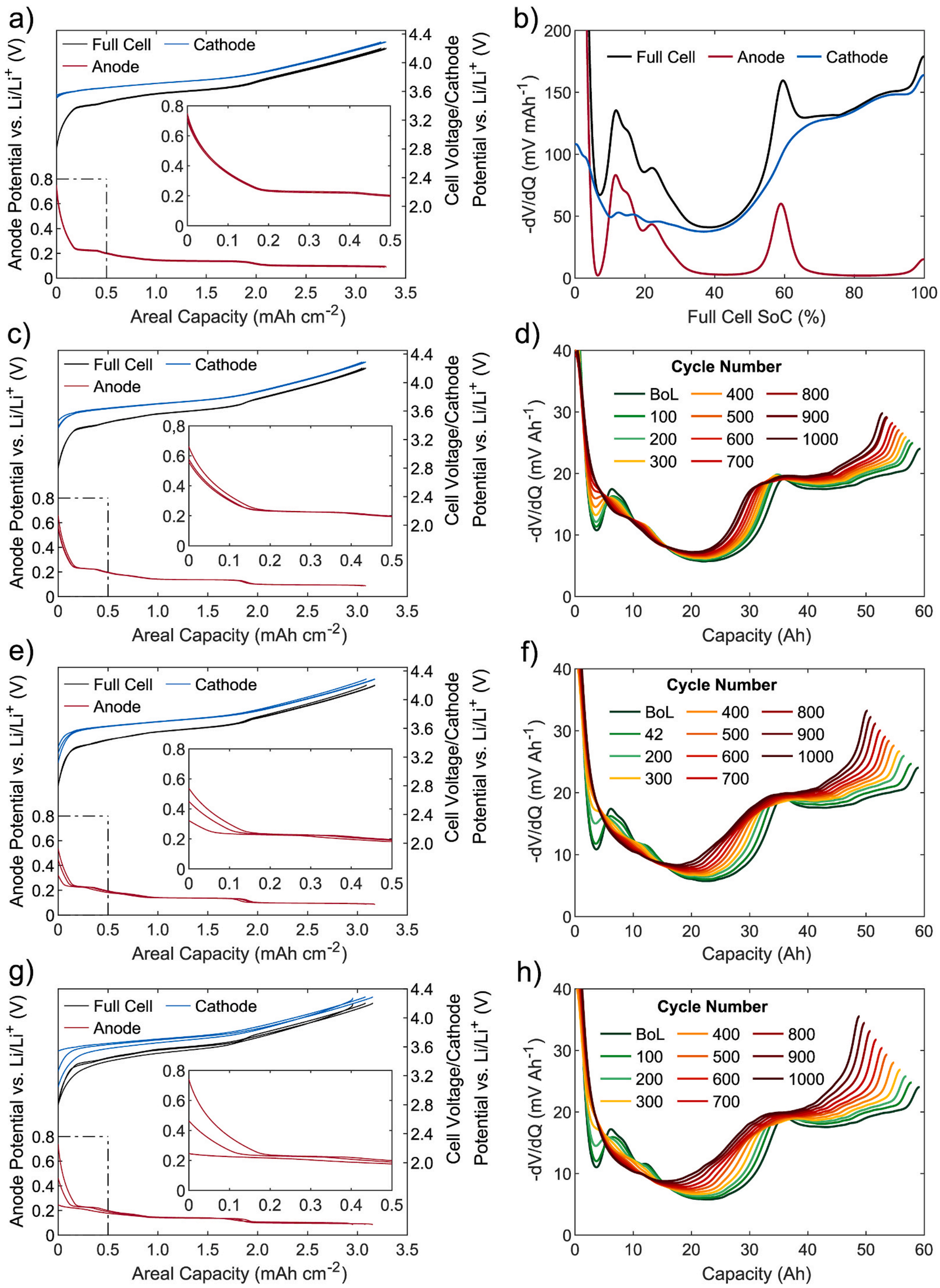

Fig. 5. Cell voltage, anode and cathode potentials from three-electrode full-cell discharge measurements (I = C/20) from harvested electrodes of a) a fresh cell (SoH 100%), c) cell 1 (SoH 89.6%), e) cell 2 (SoH 85.5%) and g) cell 6 (SoH 83.3%). Constructed DV curve from anode and cathode potentials of the a) fresh cell and the corresponding DV curves from pouch-cell discharge measurements (I = C/10) and their evolution during aging for d) cell 1, f) cell 2 and h) cell 6.

further analyzed using ImageJ. Again, the SEM micrographs in Fig. 7 are sorted with decreasing SoH from top to bottom. The analysis of the cathode cross-section (see Fig. 7a, d, g, j) reveals both an increasing number of uncropped particles and a decreasing median size of these particles in the analyzed image section with declining cathode capacity (*cf.* Table 6). This change of cathode morphology during aging strongly suggests an increasing degree of particle cracking and pulverization of the active material accelerated with growing pressure.

The anode SEM cross-sections (*cf.* Fig. 7b, e, h, k) on the other hand exhibit no apparent change in morphology. The upper and lower electrode coatings were examined separately and a mean value for the porosity was calculated (see Table 6). At BoL there is a difference of 2.9% between the porosity of the upper and the lower coating. Therefore any differences measured for the aged anodes are assumed to be random and due to a variance of cell parameters. It should be noted that the crack along the current collector (*cf.* Fig. 7k) probably formed during sample preparation and has no influence on the porosity analysis, as both layers were analyzed separately. The top-view micrographs show a similar amount of anode surface depositions for all aged cells. However, the depositions on the anode of cell 6 were non-uniform, as some areas exhibited a significantly lower amount of surface deposits (see Fig. S2) and other areas a thick surface film layer covering all graphite particles

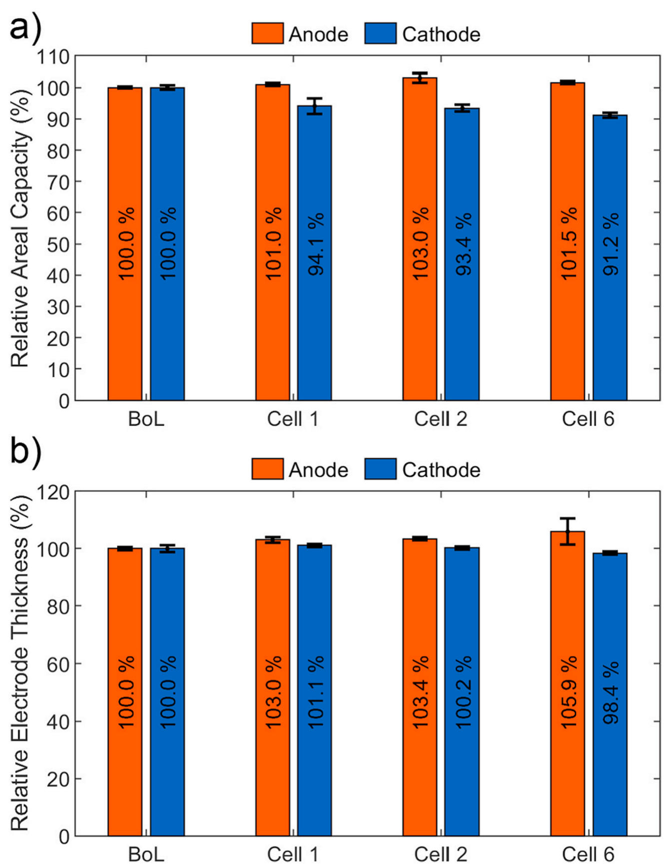

Fig. 6. a) Normalized discharge capacities of half-cells built from a fresh cell, cell 1, cell 2 and cell 6. b) Relative electrode thickness measured at 6 points per electrode.

(*cf.* Fig. S3).

Cell 4 (SoH 81.2%) was fully charged prior to the cell opening in order to evaluate the homogeneity of lithium distribution. No electrodes were harvested from cell 4 as the cell opening was performed at atmosphere. The fully charged anode of cell 4 displays brown surface deposits along the upper and lower electrode edge (*cf.* Fig. 7m). During cell opening, some of these surface film layers visibly reacted with the ambient atmosphere leading to heat generation. This suggests a high lithium content of the surface film, which is presumed to be plated metallic lithium. In addition, the uneven color distribution shows an inhomogeneous lithiation of the graphite at the edge of the anode and especially in the corners of the electrode stack. Both aging mechanisms are in accordance with literature for cells cycled under pressure [15,16,28]. It should be noted that the white depositions on the surface stem from frozen ambient humidity, as the frozen cell (ca. −75 °C) was opened at atmosphere.

Our results are mostly in agreement with literature. Cannarella et al. [15] reported increasing inhomogeneity of lithium distribution and increasing amounts of lithium deposition for cells cycled under higher pressure [15,28]. The reasons for this might be local pressure peaks resulting in a compression of the separator, leading to locally limited diffusion. Higher current densities occur around this defect, promoting local lithium plating [20–22,57]. Cannarella et al. [15] reported no active material loss of the LCO cathode even at pressures around 2.5 MPa. This agrees with Louli et al. [5] who investigated pressure evolution and capacity fade for NMC 532, NCA and LCO-based cells and found no evidence of active material loss. Mussa et al. [58] reported an influence of constant pressure on active lithium loss but neither on anode nor cathode active material loss. In contrast, Wünsch et al. [31] reported cathode active material loss due to particle cracking for a rigid braced graphite/NMC 622 cell based on EIS-measurements but did not perform a post-mortem analysis to prove this. To the authors' knowledge, no evidence of pressure induced cathode aging has been reported in literature so far. This may be due to the fact that most aging studies under pressure were conducted either under constant pressure [58], with lower setup stiffnesses and therefore lower pressures [30,31], with other cathode chemistries that are not as prone to particle cracking as Ni-rich NMC [15,22,30,58] or investigated significantly less cycles [5,19,24,29].

3.4. Aging model

A common approach to model mechanical degradation due to a cyclic load is to employ a material fatigue model according to the Wöhler curve, which is based on the assumption that a material can withstand a specific number of cycles until failure. The following model equations are taken from Laresgoiti et al. [59], who developed a model for cyclic SEI cracking. According to the empirical Basquin law of fatigue (eq. (2)), the number of cycles to failure N_{max} depends on the stress amplitude σ_{ampl} of the cyclic load, the static yield strength σ_{yield} and the slope of the respective Wöhler curve m, which are material parameters and commonly derived from experiments [59–61].

$$N_{max} = \left(\frac{\sigma_{yield}}{\sigma_{ampl}}\right)^{\frac{1}{m}} \qquad \text{(eq. 2)}$$

The stress amplitude σ_{ampl} is calculated from the maximum and minimum stress σ_{max} and σ_{min} (eq. (3)).

$$\sigma_{ampl} = \frac{\sigma_{max} - \sigma_{min}}{2} \qquad \text{(eq. 3)}$$

According to the Palmgren-Miner [62,63] rule, the cumulative damage D caused by k independent stress events with n_i load cycles can be linearly accumulated (eq. (4)).

$$D = \sum_{i=1}^{k} \frac{n_i\,(\sigma_{ampl,i})}{N_{max,i}} = \sum_{i=1}^{k} \left(\frac{\sigma_{ampl,i}}{\sigma_{Yield}}\right)^{\frac{1}{m}} n_i\, \sigma_{ampl,i} \qquad \text{(eq. 4)}$$

Laresgoiti et al. [59] assumed uniform external stress during cyclic aging and further simplified eq. (4). In the context of pressure evolution due to cell swelling, this is obviously not the case and therefore eq. (4) was used to model the degradation for each cycle (eq. (5)). As every load cycle appears only once, n_i was removed from the equation.

$$D(k) = \sigma_{Yield} \sum_{i=1}^{k} \sigma_{ampl,i}^{\frac{1+m}{m}} \qquad \text{(eq. 5)}$$

Assuming a proportional capacity loss from the mechanical damage per cycle, a SoH-model is proposed and further simplified (eq. (6)).

$$SOH = 100\ \% - a\,\sigma_{Yield} \sum_{i=1}^{k} \sigma_{ampl,i}^{\frac{1+m}{m}} = 100\ \% - b \sum_{i=1}^{k} \sigma_{ampl,i}^{\ c} \qquad \text{(eq. 6)}$$

Simultaneous linear regression fitting for all cell tests determined an optimal fit for b = 0.0276 and c = 0.5954. Cell test results and the SOH-model are depicted in Fig. 8 with the respective R^2 and RMSE for each cell. The aging model shows a good fit and a low RMSE for all cells except cell 5, where the maximum pressure at EoT was too low (~0.2 MPa) to cause any significant particle cracking. In fact, the DV curves of cell 5 (*cf.* Fig. S3) confirm lithium loss presumably from SEI-growth as the dominant aging mechanism. This shows an important limitation of this aging model, as it incorporates only one aging mechanism. Additionally, this model fails to predict a decreasing aging trend, *e.g.* for cell 4, where the maximum pressure of the test setup was reached, which is believed to be the cause for this aging trend.

Overall, the aging model is able to describe the effects of pressure on capacity loss in good agreement with the data. In order to employ this model to predict the lifetime, it needs to be coupled with other aging models that incorporate different stress factors, *i.e.* temperature,

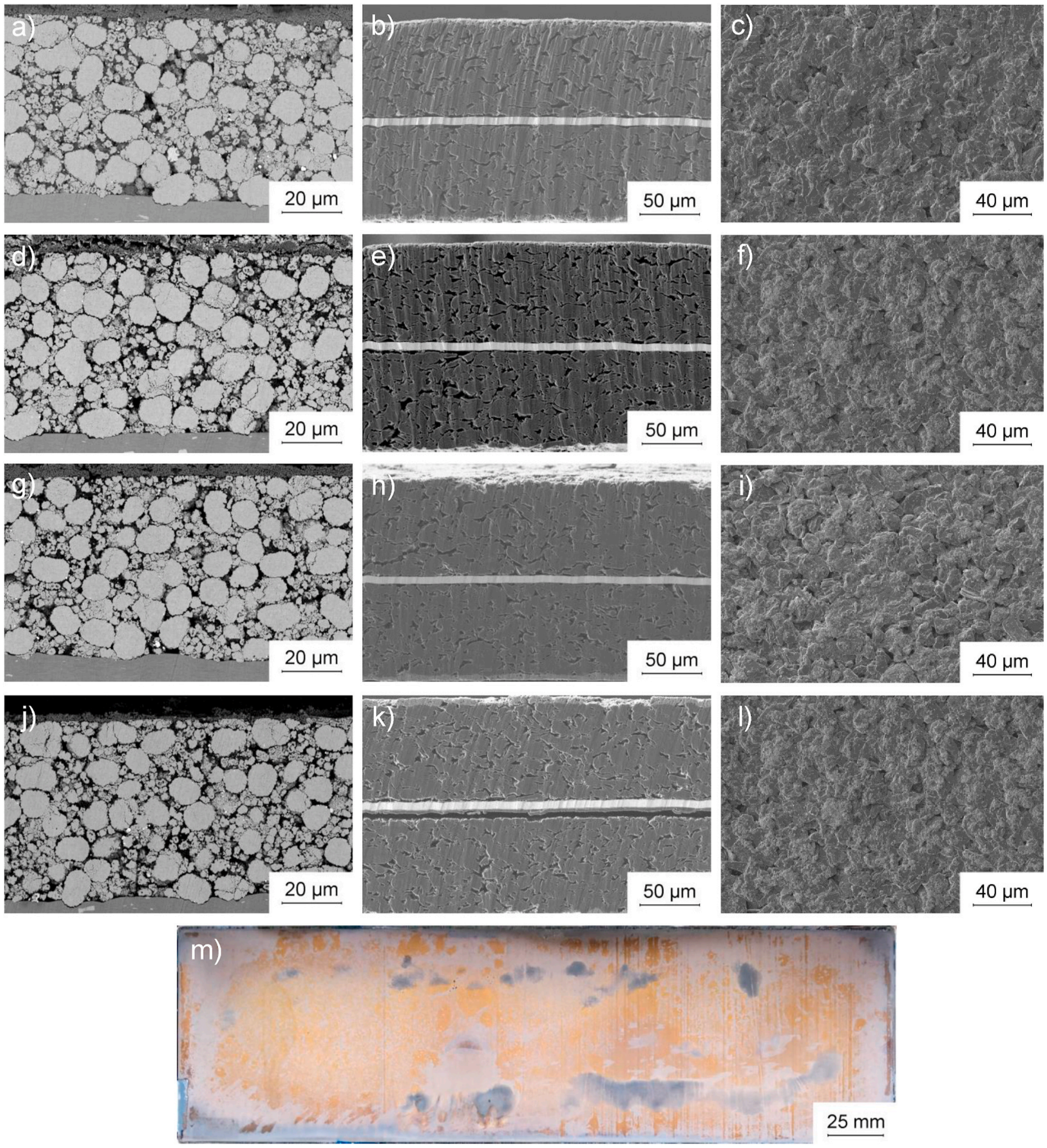

Fig. 7. SEM micrographs of cathode (left) and anode (middle) cross-sections as well as anode top view (right) of a)-c) a fresh cell (SoH 100%), d)-f) cell 1 (SoH 89.6%), g)-i) Cell 2 (SoH 85.5%) and j)-l) Cell 6 (SoH 83.3%). m) Top view photography of a deep frozen, fully charged anode of cell 4 (SoH 81.2%).

currents and DoD. The proposed model can describe the effects of pressure on the capacity loss due to particle cracking and thus is able to predict, if the pressure evolution can be predicted and modeled accordingly, *i.e.* by determining a constant growth rate [16] and employing a coupled mechanical model [17,34,35].

4. Conclusion

In this work, we have analyzed the effects of module design, *i.e.* module stiffness and initial compression, on capacity loss, internal resistance, pressure evolution and cell swelling using a central composite experimental design. A long-term aging study was conducted with large-format automotive-grade lithium-ion cells (graphite/NMC 622) in actively controlled tests setups, capable of mechanically simulating the desired module bracing.

Statistical analysis reveals significant linear, quadratic and interaction effects of module stiffness and initial compression on maximum pressure, SoH and internal resistance after 1000 cycles. A high module stiffness and a high initial compression lead to the strongest pressure evolution, highest capacity loss and increase in internal resistance. Correlation analysis shows a strong linear correlation of maximum pressure and state of health, internal resistance as well as energy density.

A subsequent post-mortem analysis reveals an increasing cathode active material loss with increasing pressure. In cross-section SEM micrographs, cracking of the NMC particles is observed, leading to fractures and smaller particles, exposing new surfaces to the electrolyte. Three-electrode measurements and the pouch-cell DV curves reveal increasing inhomogeneity, *i.e.* inhomogeneous cathode active material loss, with increasing pressure as main degradation mechanism. A cell opened at 100% SoC revealed an inhomogeneous lithiation of the anode and surface films, which are assumed to be metallic lithium deposits due to lithium plating.

Based on the results of post-mortem analysis, a semi-empirical aging model is proposed that links pressure evolution to capacity loss using a fatigue-based approach. This aging model can generate valuable insights for optimizing module design. In order to employ this model for lifetime prediction, it needs to be coupled with at least two other models: a general aging model, predicting the lifetime based on stress factors such as temperature, currents or DoD and a force prediction model to enable extrapolation. Additionally, the effects and the corresponding aging

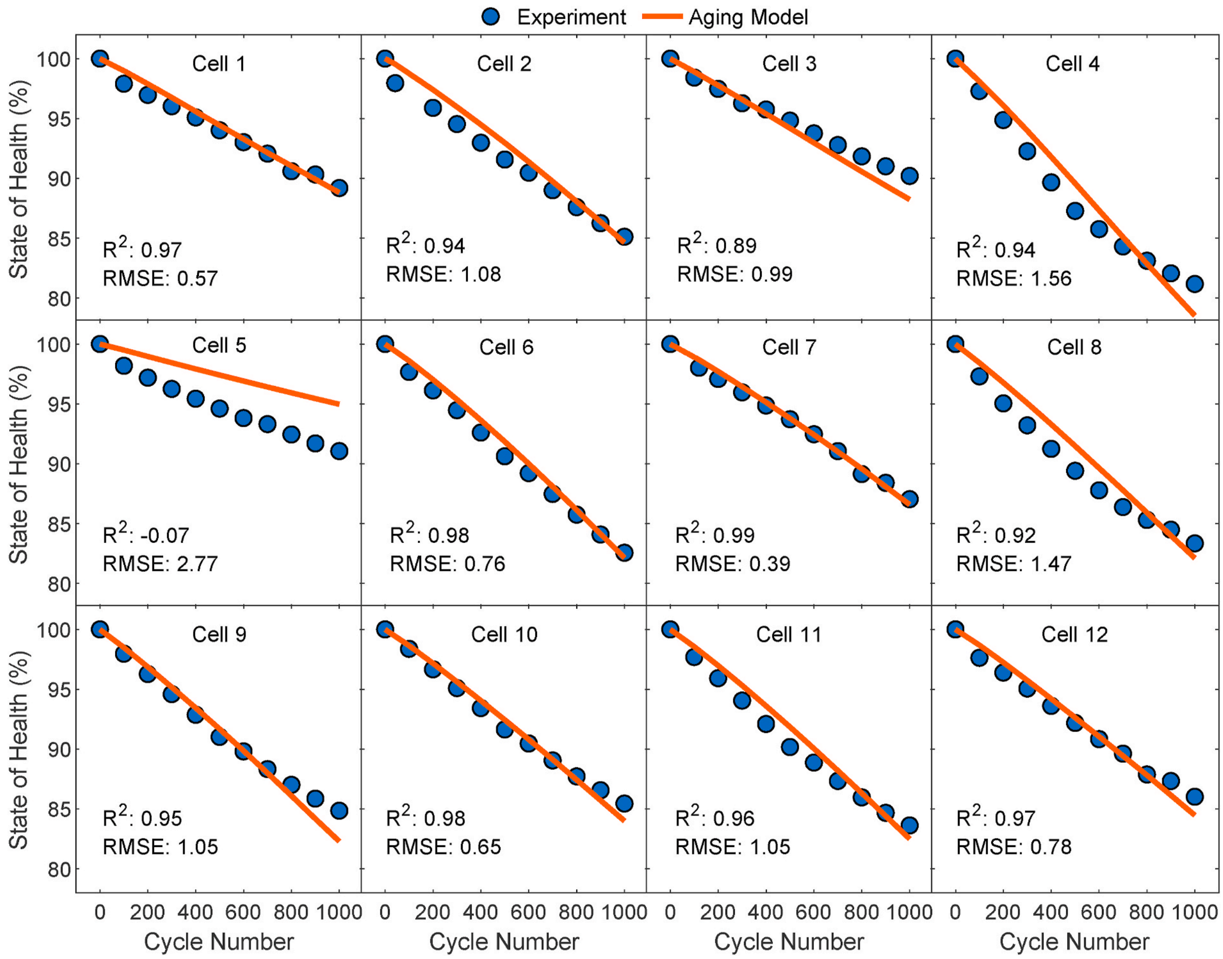

Fig. 8. Fatigue-based aging model versus state of health obtained from check-ups for each cell with the corresponding R^2 and RMSE values.

model need to be further validated under more realistic aging conditions such as driving cycles and at different temperatures. In the future the influence of different cell types and chemistries as well as form factors on cell swelling and pressure evolution should be investigated.

CRediT authorship contribution statement

Tobias Deich: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Mathias Storch:** Methodology, Investigation, Writing – review & editing, Visualization. **Kai Steiner:** Supervision, Writing – review & editing. **Andreas Bund:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2021.230163.

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